

of the main system when a new PSU is introduced.

Duplicated control unit based FT systems. When the system has to be completely fault-tolerant, duplication beyond the PSU is needed as other faults may still cause problems despite having duplicated PSUs in place. In the duplicated controller architecture, CPUs are duplicated so that, if one CPU or associated logic fails, the other CPU takes over. Duplicated CPU based FT systems are based on a combination of hardware and software.

The FT mechanism works on two essential features that the processors have—watchdog timer and high-

speed serial or parallel link between two CPUs. Fig. 3 shows how duplicated CPU based FT systems are implemented.

When an FT system based on duplicated CPUs is implemented, the following three aspects need to be understood well:

1. Time taken for the good CPU to take over from the faulty one (known as switch-over time)
2. Consistency of system data and user data between the two CPUs (data integrity)
3. Interface to the common control element by the two CPUs (redundant CPU bus interface)

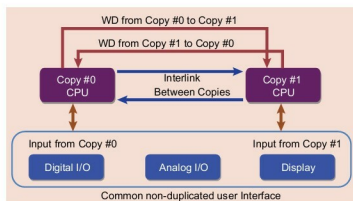


Fig. 3: How duplicated CPU based FT systems are implemented

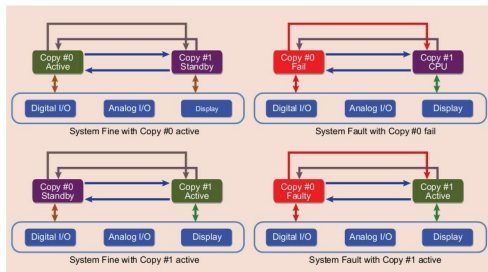


Fig. 4: Sequence of events

Watchdog timer

Watchdog timer (WDT), sometimes called a watchdog, is an electronic timer that is used to detect and recover from CPU malfunctions. During normal operation, the CPU regularly restarts the WDT to prevent it from elapsing or timing-out. If, due to a hardware fault or program error the processor fails to restart the watchdog, the timer will elapse and generate a time-out signal or an interrupt to initiate corrective action.

implemented to help us understand the FT operation well.

Duplicated CPU based system uses two CPU circuit boards (which are identical) with two types of interconnections between them. Each CPU has a signal called watchdog out (WD OUT) and watchdog in (WD IN), and high-speed interlink in (HS-ILNK-IN) and high-speed interlink out (HS-ILNK-OUT). Since the system has only duplicated controllers, these are connected to common functions that they control (Fig. 3).

These common functions receive control and data from both CPUs. Hence, the common function should have the facility to be controlled by either CPU. Also, if an FT system has to be implemented, it has to be done from the concept phase covering both hardware and software.

How a duplicated CPU FT system works. When the system starts, both CPUs are good and either of them can control the system. This is implemented by a hardware based dice mechanism, which is similar to a coin-toss function. At the start, the dice circuit will randomly make one CPU active.

Typically, in a duplicated CPU FT system, CPUs are called copy #0 and copy #1. So, at the start itself, one of them is actively controlling the system. Both CPUs will be punching the watchdog, so that no time-out occurs. At the same time, the system software through the high-speed interlinks updates all critical data on a run-time basis, so that both CPUs are in identical states. This mode is called duplex mode.

When the active copy develops a fault (Fig. 4), it fails to punch the watchdog and a time-out occurs. This triggers a signal to the other copy, which straightaway takes control of the system and raises an alarm, indicating that the CPU switch-over has happened. This mode is called simplex mode.

In the meanwhile, the faulty CPU with a watchdog timer restarts and runs self-diagnostics to iden-